

Homework 9

Connor Mooneyhan

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Disclaimer: AI was used for some question clarification and for some LaTeX help, as well as for giving a hint on the forward direction of problem 2.

1. Let R be a commutative ring, $a \in R$, and $f_1, \dots, f_r \in R[x]$.

(a) Prove the equality of ideals in $R[x]$: $\langle f_1, \dots, f_r, x - a \rangle = \langle f_1(a), \dots, f_r(a), x - a \rangle$.

(b) Prove the useful substitution trick: $\frac{R[x]}{\langle f_1, \dots, f_r, x - a \rangle} \cong \frac{R}{\langle f_1(a), \dots, f_r(a) \rangle}$.

Proof.

(a) Let $I = \langle f_1, \dots, f_r, x - a \rangle$ and $J = \langle f_1(a), \dots, f_r(a), x - a \rangle$. It will suffice to show that each generator of I is in J and vice versa. Since $x - a$ is a generator of both I and J , we can focus on the other generators.

First, we show that $f_i \in J$ for $1 \leq i \leq r$. If f_i is a constant polynomial, then $f_i = f_i(a)$, so inclusion is obvious. Taking f_i to be nonconstant then, we use the division algorithm to obtain $q_i, r_i \in R[x]$ such that $f_i = q_i(x - a) + r_i$. If $r_i = 0$, then $f_i = q_i(x - a) \in J$. If $r_i \neq 0$, then $\deg r_i < \deg(x - a)$, so r_i is constant. Taking this together with the fact that $f_i(a) = r_i(a) = r_i$, we conclude that $f_i \in J$ since it is the sum of elements of J , namely $q_i(x - a)$ and $r_i = f_i(a)$.

Now we want to show that $f_i(a) \in I$ when $1 \leq i \leq r$. Using the same notation and facts established above, we can write $f_i = q_i(x - a) + r_i$, so transforming the equation to become $f_i - q_i(x - a) = r_i$ tells us that $f_i(a)$ is a sum of elements in I and is thus in I .

(b) Define $S = \frac{R}{\langle f_1(a), \dots, f_r(a) \rangle}$, and let $\varphi : R[x] \rightarrow S$ be defined by $f \mapsto f(a)S$. The image of φ is $R/\langle f_1(a), \dots, f_r(a) \rangle$ since given any $b \in R$, the constant polynomial $b \in R[x]$ maps to $b\langle f_1(a), \dots, f_r(a) \rangle$. The kernel of φ is $\langle f_1, \dots, f_r, x - a \rangle$ by construction, so we get the desired isomorphism from the First Isomorphism Theorem.

□

2. A commutative ring R is said to be *local* if there is a unique maximal ideal of R . Show that a ring R is local if and only if the set of elements of R that are not invertible is an ideal of R .

Proof. Let U be the set of elements of R that are not invertible. It is useful to show that every proper ideal of R is a subset of U . Indeed, if a proper ideal I has an invertible element x , then $x^{-1}x = 1 \in I$, so $I = R$. Thus every element of a proper ideal is noninvertible. Also useful to know is that U always satisfies absorption, since if a is noninvertible, $b \in R$, and ab has an inverse c , then $1 = (ab)c = a(bc)$, so bc would be an inverse of a , contradicting noninvertibility. U also always contains 0 unless R is the trivial ring (in which case $U = R$ so U is an ideal), and it is closed under taking additive inverses. Thus the only way U can fail to be an ideal is to not be closed under addition.

($\implies$) If R is local, then given $x, y \in U$, if $x+y$ is invertible, then $R = (x+y) = (x) + (y)$. Since (x) and (y) are both proper because x and y are noninvertible, each is contained in the unique maximal ideal of R , but $x+y$ is not. This is a contradiction, so we must have that U is closed under addition and is thus an ideal.

($\impliedby$) If U is an ideal, then given an ideal I of R , either $I = R$ or $I \subset U$, so U is maximal. Indeed, if I is a maximal ideal, then the fact that $I \subset U$ implies $I = U$, so U is the unique maximal ideal of R . $\square$

3. Let $\varphi : R \rightarrow S$ be a homomorphism of commutative rings, and suppose $\mathfrak{p}$ is a prime ideal of S . Prove that $\varphi^{-1}(\mathfrak{p})$ is a prime ideal of R .

Proof. First we show that $\varphi^{-1}(\mathfrak{p})$ is an ideal of R . Let $a \in R$ and $b \in \varphi^{-1}(\mathfrak{p})$. Then $\varphi(ab) = \varphi(a)\varphi(b)$, and since $\varphi(b) \in \mathfrak{p}$, $\varphi(ab) \in \mathfrak{p}$ by absorption, and thus $ab \in \varphi^{-1}(\mathfrak{p})$.

Now given any $a, b \in R$ such that $ab \in \varphi^{-1}(\mathfrak{p})$, $\varphi(ab) = \varphi(a)\varphi(b) \in \mathfrak{p}$, and thus either $\varphi(a) \in \mathfrak{p}$ or $\varphi(b) \in \mathfrak{p}$. WLOG, assume $\varphi(a) \in \mathfrak{p}$. Then $a \in \varphi^{-1}(\mathfrak{p})$, so applying the same argument to the “ b ” case, we see that $\varphi^{-1}(\mathfrak{p})$ is prime. $\square$

4. For a commutative ring R , let $S^{-1}R$ be the ring of fractions as defined on the previous homework set.

- (a) Let $\mathfrak{p}$ be a prime ideal of R , and let $S = R \setminus \mathfrak{p}$, that is, the complement of $\mathfrak{p}$ in R . Prove that S is a multiplicatively closed subset of R .
- (b) Let S be as in the previous part. Show that $S^{-1}R$ is a local ring, with maximal ideal $\{p/s : p \in \mathfrak{p}, s \in S\}$. This ring is denoted $R_{\mathfrak{p}}$ in the literature, and is called “ R localized at $\mathfrak{p}$ ”.
- (c) Realize the ring $\mathbb{Z}_{(2)}$ as a subset of $\mathbb{Q}$, and show that it is a subring under the usual operations. Identify the maximal ideal of $\mathbb{Z}_{(2)}$ as a subset of $\mathbb{Q}$ as well.

Proof.

- (a) Let $a, b \in S$. If $ab \notin S$, then $ab \in \mathfrak{p}$, so either a or b is in $\mathfrak{p}$ since $\mathfrak{p}$ is prime. This contradicts a and b both being in S , so $ab \in S$.
- (b) Let $I = \{p/s : p \in \mathfrak{p}, s \in S\}$. We first show that I is an ideal. It is closed under addition since if $p_1, p_2 \in \mathfrak{p}$ and $s_1, s_2 \in S$,

$$\frac{p_1}{s_1} + \frac{p_2}{s_2} = \frac{p_1 s_2 + p_2 s_1}{s_1 s_2} \in I$$

because $p_1 s_2, p_2 s_1 \in \mathfrak{p}$ since $\mathfrak{p}$ is an ideal. Also, if $r/s \in S^{-1}R$, then for $p_1 \in \mathfrak{p}$ and $s_1 \in S$,

$$\frac{r p_1}{s s_1} = \frac{r p_1}{s s_1} \in I$$

since $r p_1 \in \mathfrak{p}$. Thus I is an ideal.

Now if J is any ideal containing I where $I \neq J$, then we can choose an element $r/s \in J$ such that $r/s \notin I$, and thus $r \notin \mathfrak{p}$, so $r \in S$. Then r/s is invertible because

$$\frac{r s}{s r} = \frac{r s}{s r} = \frac{r s}{r s} \sim \frac{1}{1}$$

since $1 \cdot (rs - rs) = 0$. Thus $J = S^{-1}R$ as desired, and I is maximal.

- (c) $\mathbb{Z}_{(2)}$ is the set of rationals whose reduced form has an odd denominator. Given two elements $r_1/s_1, r_2/s_2 \in \mathbb{Z}_{(2)}$, the denominator of both their sum and their product is $s_1 s_2$, which is odd since both s_1 and s_2 are odd, and thus $\mathbb{Z}_{(2)}$ is closed under addition and multiplication. Also, the identities $0/1$ and $1/1$ are both in $\mathbb{Z}_{(2)}$, so indeed it is a subring.

□

5. Let k be an algebraically closed field, and let I be an ideal of $k[x]$. Prove that I is maximal if and only if $I = \langle x - c \rangle$ for some $c \in k$.

Proof. Since k is a field, we know that $I = \langle f \rangle$ where f is the unique monic polynomial of least degree in I . Since any $g \in k[x]$ has a root c_g , we have $g = (x - c_g)q_g$ for some $q_g \in k[x]$, and we could repeat this process (starting now with q_g) until we have expressed g as a product of monic degree 1 polynomials.

Since I is maximal and k is a field, f is irreducible. The only irreducible elements of $k[x]$ are of degree 1, so f must have degree 1. Since it is also monic, f is of the form $x - c$ for some $c \in k$ as desired.

□

6. Explicitly construct a field of order 8, 27, and 16. Be careful with the one of order 16!

Proof.

1. The field

$$\frac{\mathbb{F}_2}{\langle x^3 + x + 1 \rangle}$$

has order $8 = 2^3$ since $x^3 + x + 1$ is irreducible over $\mathbb{F}_2$ and has degree 3.

2. The field

$$\frac{\mathbb{F}_3}{\langle x^3 + x^2 + x + 2 \rangle}$$

has order $27 = 3^3$ since $x^3 + x^2 + x + 2$ is irreducible over $\mathbb{F}_3$ and has degree 3.

3. The field

$$\frac{\mathbb{F}_2}{\langle x^4 + x + 1 \rangle}$$

has order $16 = 2^4$ since $x^4 + x + 1$ is irreducible over $\mathbb{F}_2$ and has degree 4.

□